



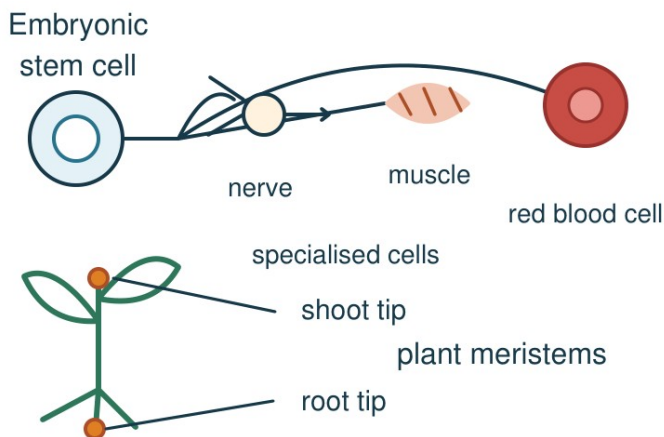
Growth and stem cells

Today I can... ☐ I can distinguish making more cells from cells taking on a specialised job. ☐ I can compare animal growth with plant growth. ☐ I can identify a role for an animal stem cell or a plant meristem.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____

Arrival

Look at this week's seedling and skin-repair model. Point to one detail you notice.



My response

Starter

Recall three ideas from earlier lessons. Show one answer at a time.

1. What two matching products follow body-cell mitosis and cell splitting?

2. Which process moves particles from low to high concentration using energy?

3. In which part of a eukaryotic animal cell are chromosomes found?



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Talk-then-sort

Complete the missing step in the part-done model. Speak, then match the cards.

Skin: mitosis adds cells → differentiation gives a job. Shoot: meristem adds cells → _____ lengthens cells → differentiation.

Choices: differentiation · division · elongation

Card or step	My group or match
Makes more cells	
Plant cells become longer	
Cell gains a tissue job	

A skin model shows division. What gives a new cell its job?

Independent F

Place divide, elongate and differentiate cards on the skin/shoot branches; omit elongate on the skin branch.

In the shoot, first cells ...; then they ...

My response

Independent M

Explain why mitosis alone does not give a new cell its specialised tissue job.

More cells are made by ..., but a job develops when ...

My response



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Independent H

Compare growth in animal tissue with a plant shoot, including an appropriate stem-cell region and what each process adds.

Both organisms ...; however, the plant also ... because ...

My response

Go deeper

A tissue gains more cells but no new cell types. What does the model explain, and what is missing?

My response

Correct a diagram labelling new skin cells “stem cell”; justify two changes.

My response

Exit

Show what you know with a model and explain one reason.

My exit response

My feedback was ... Name what you heard at Review.

Next I need to ... Add the step showing how one newly made cell takes on a specialised role.

Word help

Word	Meaning	Example	My language
differentiation	a cell becoming specialised for a particular job	A developing skin cell differentiates to do a job in skin.	
stem cell	a cell that can divide and form specialised cell types	Some adult stem cells make replacement blood cells.	
meristem	a plant region whose cells divide to make new cells	A root tip contains a meristem.	
growth	an increase in size, including by making more cells	A shoot grows as its cells divide and elongate.	